

Module 4: Supervised Learning
Submodule: 4.4 Generalized Linear Models
Lecture: Generalized Linear Models (GLM)

XCS229 Machine Learning

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I. Introduction

In this lecture, we discuss the connection between inference and hypothesis functions, linking them to **generalized linear models (GLMs)**. We aim to clarify how these concepts relate to our previous work with models such as logistic regression.

II. Structure of Generalized Linear Models

In this section, we will describe a method for constructing GLM models for problems such as these.

III. Design Choices and Assumptions

Here, we will clarify how design choices impact model creation. More generally, consider a classification or regression problem where we would like to predict the value of some random variable y as a function of x . To derive a GLM for this problem, we will make the following three assumptions (and these are not necessarily right or wrong) about the conditional distribution of y given x and about our model:

1. $y | x; \theta \sim \text{ExponentialFamily}(\eta)$. ($\sim$ means “behaves according to...”)
Given x and θ , the distribution of y follows some exponential family distribution, with parameter η .

Examples include:

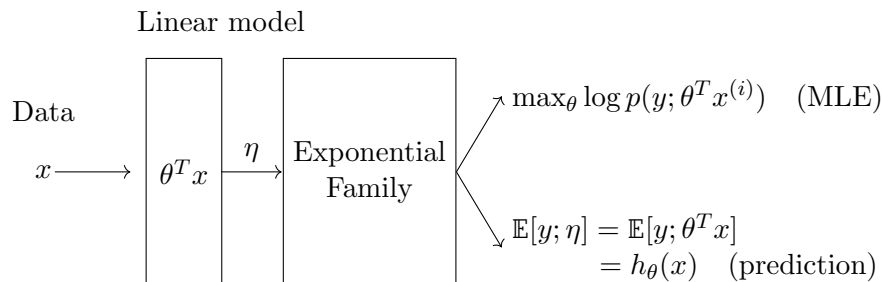
- **Binary** $y \rightarrow$ Bernoulli distribution (logistic regression)
- **Real** $y \rightarrow$ Gaussian distribution (linear regression)
- **Count data** $y \rightarrow$ Poisson distribution
- **Positive data** $y \rightarrow$ Gamma or Exponential distribution
- **Infinite dimensional crazy distribution structure** $y \rightarrow$ Dirichlet distribution

2. The natural parameter η and the inputs x are related linearly:

$$\eta = \theta^T x \quad \text{where} \quad \theta \in \mathbb{R}^d, x \in \mathbb{R}^d$$

(Or, if η is vector-valued, then $\eta_i = \theta_i^T x$.)

3. Given x , our goal is to predict the expected value of $T(y)$ given x .
In most of our examples, we will have $T(y) = y$, so this means we would like the prediction $h(x)$ output by our learned hypothesis h to satisfy $h(x) = \mathbb{E}[y|x; \theta]$. (Note that this assumption is satisfied in the choices for $h_\theta(x)$ for both logistic regression and linear regression. For instance, in logistic regression, we had $h_\theta(x) = p(y = 1|x; \theta) = 0 \times p(y = 0|x; \theta) + 1 \times p(y = 1|x; \theta) = \mathbb{E}[y|x; \theta]$.)



IV. Terminology in Generalized Linear Models

In this section, we will explore important terminology and concepts related to GLMs. We will discuss model parameters, natural parameters, canonical parameters and link functions. Furthermore, we will understand the flow of these concepts through examples of logistic and linear regression.

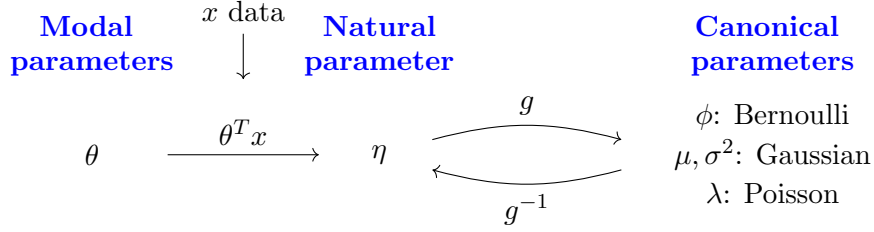
1. **Model Parameters (θ):** Model parameters, represented by θ , are the coefficients that we estimate during the training of linear and logistic regression models. These are what we train on when we fit a model to our data. The model takes incoming data x and processes it through the equation $\theta^T x$, allowing us to make predictions based on inputs and learned weights.
2. **Natural Parameter (η):** The natural parameter, denoted as η , is derived from the model parameters. Specifically, in generalized linear models, we define it as $\eta = g(\theta^T x)$, where g is a link function that transforms the linear combination of features into a suitable format for modeling the response variable.
3. **Canonical Parameters:** Canonical parameters refer to specific parameters for common probability distributions:
 - **Bernoulli:** For a Bernoulli distribution, the canonical parameter is ϕ , representing the probability of success.
 - **Gaussian:** For the Gaussian distribution, the canonical parameters are μ (mean) and σ^2 (variance).
 - **Poisson:** For the Poisson distribution, the canonical parameter is λ , representing the rate of occurrence.

The **link function** g is used to connect the model's linear predictor ($\theta^T x$) to the expected value of the response variable. Its role is critical in relating model parameters to canonical parameters. There is also an inverse link function g^{-1} , which transforms back from the natural parameter η to the canonical parameters (ϕ, μ, λ).

The function g giving the distribution's mean as a function of the natural parameter

$$g(\eta) = \mathbb{E}[T(y); \eta]$$

is called the **canonical response function**. Its inverse, g^{-1} , is called the **canonical link function**. Thus, the canonical response function for the Gaussian family is just the identity function; and the canonical response function for the Bernoulli is the logistic function.



Flow of Concepts

Now, we will discuss the flow of concepts, illustrating how we move from model parameters to natural parameters and then to canonical parameters.

1. Model Parameters to Natural Parameters:

The journey begins with the model parameters (θ), which are estimated through training. Once we have θ , we can input our data x into the equation $\theta^T x$. This result is then fed into the link function g to yield the natural parameter η .

2. Natural Parameters to Canonical Parameters:

After obtaining η , we can apply the inverse link function g^{-1} to derive the canonical parameters. This connection allows us to understand our model in terms of well-established statistical distributions. For example, if we have derived η from our model parameters, we can use g^{-1} to find the corresponding canonical parameters like ϕ , μ , and λ .

V. Examples

Next, we will illustrate these concepts with practical examples of logistic and linear regression, demonstrating how they apply.

Logistic Regression

In the context of logistic regression, the predicted value $h_\theta(x)$ relates to the probability of an event occurring. Using the logistic link function, we can express the probability of success (a Bernoulli variable $y|x; \theta \sim \text{Bernoulli}(\phi)$) in terms of the natural parameter η . The natural parameter η serves as an intermediary that helps us establish the connection to the canonical parameter ϕ , which represents the probability.

Note that if $y|x; \theta \sim \text{Bernoulli}(\phi)$, then $\mathbb{E}[y|x; \theta] = \phi$. Furthermore, in our formulation of the Bernoulli distribution as an exponential family

distribution, we had $\phi = 1/(1 + e^{-\eta})$. So, we have

$$\begin{aligned} h_{\theta}(x) &= \mathbb{E}[y|x; \theta] \\ &= \phi \\ &= \frac{1}{1 + e^{-\eta}} \\ &= \frac{1}{1 + e^{-\theta^T x}}. \end{aligned}$$

Where $\mathbb{E}[y|x; \theta] \in [0, 1]$.

How do we use this for classification?

$$\begin{aligned} h_{\theta}(x) > 0.5 &\implies 1 \\ \text{otherwise} &\implies 0. \end{aligned}$$

Linear Regression

In linear regression, the model's output is directly $\theta^T x$, which corresponds to the mean of the underlying distribution (Gaussian $\mathcal{N}(\mu, \sigma^2)$). This means the natural parameters and model parameters align. Thus, while we operate with model parameters during training, we simultaneously understand them as representing the mean of our Gaussian distribution.

As we saw previously, in the formulation of the Gaussian as an exponential family distribution, we had $\mu = \eta$. So, we have:

$$\begin{aligned} h_{\theta}(x) &= \mathbb{E}[y|x; \theta] \\ &= \mu \\ &= \eta \\ &= \theta^T x. \end{aligned}$$

VI. Summary

In this section, we will summarize the key takeaways from the lecture.

- Generalized linear models (GLMs) offer a flexible framework for modeling various types of response variables, including binary, continuous and count data.
- The structure of a GLM involves defining model parameters (θ), natural parameters (η), and canonical parameters, which relate to specific probability distributions.
- Understanding the link function is crucial, as it connects the linear predictor to the expected value of the response variable.

- The flow of concepts starts from model parameters, transitions to natural parameters, and then to canonical parameters, illustrating the relationships between these components.
- Practical examples, such as logistic and linear regression, demonstrate how these theoretical ideas are applied in real-world scenarios for classification and prediction tasks.

In conclusion, the generalized linear model provides a robust framework for linking various statistical properties with model parameters. By defining relationships through natural parameters and canonical parameters, inference becomes a clear process.